DIVISION ALGORITHM

Division algorithm

- Is an effective method for producing such quotient and remainder.
- There are numbers of algorithm such as decimal notation of integers, long division which provides an efficient division algorithm.
- The integer division algorithm is an important procedure for other algorithm, such as Euclidian algorithm for determining the greatest common divisor of any pair of integers.

EUCLIDEAN ALGORITHM : $a = qb + r$; $a = bq + r$

GREATEST COMMON DIVISOR

1. Determine the greatest common divisor of 372 and 132.

GREATEST COMMON DIVISOR

2. Determine the GCD of 24 and 18.

GREATEST COMMON DIVISOR

3. Solve for the GCD of 108 and 87.

LEAST COMMON MULTIPLE

1. Find the least common multiple of the following:

a. (12, 18)

b. (90, 108)

c. (24, 36, 108)